Anirudh Garg

B.Tech., Computer Science & Engineering
Indian Institute of Technology Bombay
Mail | Linkedin | Roll No: 22B1005

Examination	University/Board	Institute	Year	CPI/%
Graduation	IIT Bombay	IIT Bombay	2026	9.66
Intermediate	CBSE	Bhawan Vidyalaya Panchkula	2022	96.40%
Matriculation	CBSE	St John's High School	2020	97.00%

Scholastic Achievements

- Secured **All India Rank 50** Joint Entrance Examination **Advanced** amongst 1,50,000+ candidates *(2022)*
- Achieved **All India Rank 224** in JEE **Main** with **99.98 percentile** out of over 1 million candidates *(2022)*
- Qualified twice for Indian National Mathematics Olympiad **(INMO)** conducted by HBCSE *(2020, 2021)*
- Achieved **First Place** and included in the **Honor Roll** of American Mathematics Contest 10 **(AMC)** *(2020)*
- Recipient of National Talent Search Examination **(NTSE)** scholarship by NCERT, Govt. of India *(2020)*
- Achieved **AIR 2** amongst 1,40,000+ students in Vidyarthi Vigyan Manthan **(VVM)** by Govt. of India *(2018)*
- Ranked amongst the top 1% in Indian Olympiad Qualifier in Physics **(IOQP)** Stage-1 conducted by IAPT *(2022)*
- Bagged the **Bronze Medal** at Southeast Asian Maths Olympiad **(SEAMO)** in the Intermediate Category *(2018)*

Professional Experience

Jump Trading | *Quantitative Trader Internship* *(Summer '25)*

- Developed models for **pricing** ETFs and equities using **futures**-based adjustments and cross-asset relationships, experimenting with multiple **model architectures** and **feature selection** approaches to improve accuracy
- Validated **alpha signals** and implemented a **backtesting engine** to translate them into tradeable strategies
- Built Python tools for **hedge index selection** and **beta estimation** with accuracy matching Bloomberg benchmarks
- Acquired exposure to **market dynamics** and data-driven analysis through targeted workshops and **mock-trading**

Franklin Templeton Investments | *Software Development and Data Science Internship* *(Summer '24)*

- Automated **data extraction** from web, PDFs and workbooks using **R**, and their upload to a **Macrobond database**
- Enhanced an in-house **R** platform for investment strategies with **automated notifications**, secure login, **data change tracking** and analytical visualisations for real-time performance monitoring

Jane Street | *SEE Program* *(Winter '23)*

- Selected among the **top 50 students** from **India** invited to visit the Hong Kong office of **Jane Street Capital**, attending lectures, **mock-trading** games and research workshops on their quantitative approach
- Designed **linear regression** models to predict the values of **stock market indicators** using Stock Market data

Key Projects

Crypto Market Regime Modelling *(Winter '24)*
Quant Team | 13th InterIIT Tech Meet *Competition Project*

- Built predictive models for **BTC** and **ETH regime classification** using **Elastic Net regularised regression** and **gradient-boosted trees**, capturing nonlinear dependencies in **high-dimensional** time-series features
- Conducted statistical analysis of price–volume dynamics, **volatility** clustering, seasonality, and **cross-asset lead–lag relationships** across **OHLCV** to design **regime-shift predictors** and validate model outputs

Market Simulator *(Autumn '23)*
Prof. Ashutosh Gupta *Course Project*

- Built a multi-trader **C++** market simulation using **threads** and **sockets** to receive and process concurrent orders, applying **median trading** and **statistical arbitrage** strategies against the simulated order book
- Implemented a **custom ordered map** backed by **Red-Black trees** and **priority queues** for low-latency, lock-light handling of concurrent trade orders on the simulator's matching engine across worker threads

IRIDIUM: Static JavaScript Optimisation Framework *(Autumn '25)*
Prof. Manas Thakur *Research Project*

- Designed **IRIDIUM**, a framework for sound **static optimisation of JavaScript** that lowers source through normalisation passes into a structured **IRI** explicitly modeling bindings, environments and control flow
- Implemented **TDZ** and **alias** analyses with optimisations – **barrier removal**, **constant propagation/folding**, **dead code** and **dead binding elimination** – plus a codegen pass for Mozilla's **SpiderMonkey**; paper **accepted at OOPSLA 2026**

High-Performance ML Systems Optimization *(Spring '26)*
Prof. Mythili Vutukuru *Course Project*

- Engineered and optimized core deep learning components from scratch in **PyTorch** and **CUDA/C++**
- Developed a **Transformer** architecture and integrated an optimized **FlashAttention** kernel using custom **SRAM/HBM tiling** to minimize memory bandwidth bottlenecks and drastically lower computational overhead

Other Projects

Statistical Learning and Information Retrieval
(Spring '26)

Prof. Soumen Chakrabarti — *Course Project*

- Built and evaluated machine learning pipelines spanning **information retrieval**, **representation learning**, and **graph-based semi-supervised learning** using **PyTorch**, **PyTorch Geometric**, and **BERT**
- Developed neural ranking models and probabilistic rerankers (**MRF**), implemented **RNN/GRU** architectures from first principles, and analyzed optimization dynamics through spectral and gradient-based diagnostics
- Reimplemented the **Correct & Smooth** graph diffusion algorithm with optimized sparse GPU operations

Network Systems Performance & Traffic Analysis
(Fall '25)

Prof. Devashish Gosain — *Course Project*

- Built virtualized network testbeds using **FreeBSD**, **Linux**, and **Shadow** to study routing, traffic filtering, VPN protocols, and anonymous communication systems
- Implemented custom packet-processing modules including a **FreeBSD pf kernel firewall** and a **Wireshark dissector** for protocol fingerprinting of OpenVPN traffic independent of ports or IP addresses
- Designed benchmarking pipelines for **Tor** onion services, measuring latency, throughput, connection success, and network resilience under relay-failure stress tests

Sound JavaScript Security Analysis
(Spring '26)

Prof. Manas Thakur — *B.Tech. Project*

- Replicated Richards et al.'s 2011 **eval()** study on **10K Tranco sites** via **Playwright** + Chrome DevTools Protocol, finding 89.2% of sites use eval with 90.8% of calls potentially risky and 25.5% DOM-tainted
- Built the **Iridium Security** testbench (42 JS cases on Prototype Pollution, IPT, and Arbitrary File Access) and benchmarked **ESLint**, **Semgrep**, **ODGen** isolating **weak pointer analysis** as the shared failure mode
- Helped design **prakriti**, an **ECMA-spec-grounded** JavaScript pointer-analysis engine in **C++17** with **Copy-on-Write** graph storage (O(N) clones) and a rolling-XOR graph hash for O(1) fixpoint checks

Approximate Nearest Neighbour Search
(Summer '24)

Prof. Ajit Rajwade — *Summer Research Project*

- Collaborated on an **Amazon-funded** project to develop an algorithm for **ANN search** using **group testing**
- Implemented a **recursive method** to evaluate and split dictionary and query-vector groups based on dot-product thresholds in **large, high-dimensional datasets**, enhancing scalability and computational efficiency
- Tested **cumulative query batching** of 10K vectors, achieving results in under 2K seconds on a 60K-vector, 784-dimensional dataset

Compressed Sensing Algorithms
(Spring '24)

Prof. Ajit Rajwade | Advanced Image Processing — *Course Project*

- Implemented faster versions of the **orthogonal matching pursuit** algorithm using **Cholesky** and **QR** decompositions and the **Matrix Inversion** lemma; benchmarked them on the greyscale **Caltech256** dataset
- Compared video reconstruction using **ISTA** and **IHTA** across multiple frames on a **3-D DCT basis**

Game Theoretic Multilevel Programming
(Spring '24)

Prof. Sriram Sankaranarayanan, Prof. Avinash Bhardwaj, Prof. Swaprava Nath — *Research Project*

- Analysed the **watermelon algorithm** and its working on **price-setting problems** and **Stackelberg games**
- Worked on the intersection of the **Multilevel Critical Node problem** and **Blotto games** to model real-life defender–attacker scenarios via bilevel and trilevel optimisation formulations

sMART Optimisation
(Autumn '23)

Prof. Avinash Bhardwaj | Optimisation Models — *Course Project*

- Extended the **Continuous Facility Location problem** to find optimum placement of outlets across a city; learned **Blossom** and **Hungarian** algorithms for assignment and maximum matchings in bipartite graphs
- Linearly formulated the **Maximum Sum Eulerian Circuit** on an edge-weighted graph and discovered the longest closed path within the IITB Campus using **Python MIP** and the **CBC solver**

Positions of Responsibility

- **InterIIT Tech 14.0 Contingent Leader** for the IITB contingent, heading a team of 100 *(Sept '25 - Dec '25)*
- **Media Secretary** of the Computer Science Engineering Association, IIT Bombay *(April '23 - April '24)*
- **Teaching Assistant** for courses in **AIML**, **Logic** and **Optimization Models** *(July '24 - May '25)*
- Selected to **mentor 12 sophomores** as part of the CSE Department DAMP Team *(May '24 - April '25)*

Technical Skills

Programming Languages	C++, Python, R, SQL, Java, LaTeX, Bash, OCaml, JS
Software & Tools	Bloomberg, MATLAB, Macrobond, Slurm, Git, GDB, ChampSim, ModelSim
Libraries	NumPy, Pandas, SciPy, Matplotlib, PyTorch, Tensorflow, BeautifulSoup, Qiskit, MIP

Extracurricular Activities

- Received the **Silver Medal** at the State Level Squash Tournament conducted by the Education Dept. *(2018)*
- Secured **Third Place** at the Inter-Hostel **Street Play** and **Squash General Championships** *(2023)*